

Guidelines for Project Demonstration Video

Course: Microprocessors and Microcontrollers Laboratory (CSE 4326)

Video Type: Final Project Demonstration

Maximum Duration: 10 minutes

Video Language: English (mandatory)

Presentation Mode: Slides + Hardware Demo

Showing faces: Optional

General Instructions

1. The video **must be professional**, well-structured, and clearly narrated.
2. **The English language is mandatory.**
3. Slides can be used for **all sections except hardware demonstration.**
4. The **actual hardware must be shown live** during feature demonstration.
5. Background noise, shaky video, unreadable slides, or poor lighting should be avoided.
6. Maximum video length is **10 minutes**. Videos exceeding this limit **may lose marks**.
7. While preparing the video, the **Evaluation Rubrics** should be kept in mind, and **all required aspects should be clearly addressed**.

Evaluation Rubrics (Used for Final Grading In Project Show)

Category	Weight
Project Idea & Novelty	15%
Real-Life Application and Impact	15%
Features & Hardware Implementations	40%
Cost Effectiveness	10%
Tidiness & Professionalism	10%
Presentation	10%
Total	100%

Required Video Flow & Content

1. Introduction & Idea Description

[Slide-based presentation recommended]

Students must clearly explain:

- Project title
- Team Information
- Problem statement
- Core idea and motivation
- Why this problem is important

The idea should be explained in **simple and clear terms**, understandable to a non-expert.

2. Real-Life Applications

[Slide-based presentation recommended]

Students should describe:

- Real-life scenarios where this system can be used
- Target users (e.g., rescue teams, hospitals, industries, general public)
- How the system improves safety, efficiency, or automation

Use **realistic examples**, not exaggerated claims.

3. Core Features & System Overview

[Slide-based presentation recommended]

This section must include:

- List of **major features implemented** (minimum 3)
- Block diagram or system architecture
- Hardware components used:
 - Microcontrollers / microprocessors
 - Sensors
 - Motors / actuators
 - Power supply
- Brief explanation of **why these components were chosen**

Mention **specific model names** (e.g., Arduino Uno, ESP32, Raspberry Pi 5 etc.).

4. Feature Demonstration

Live hardware demonstration (NO slides here)

This is the **most important part** of the video.

Students must:

- Show the **actual hardware clearly**
- Demonstrate each major feature **one by one**
- Explain:
 - What input is given
 - What sensor/motor is involved
 - What output or action is produced
- Show software output if applicable (Serial Monitor, OLED, mobile app, website, etc.)

**** The demo must follow a logical flow**

**** System should appear stable and well-assembled**

Avoid skipping features or only explaining without showing. You may skip trivial features such as login/signup features, dashboard setup.

5. Costing & Budget Analysis

[Slide-based presentation recommended]

Students must show:

- Component-wise cost breakdown
- Total project cost
- Justification of cost-effectiveness
- Any alternative cheaper components (if applicable)

Emphasis should be on **performance vs cost**, not just low price.

6. Future Plans & Possible Extensions

[Slide-based presentation recommended]

Students should discuss:

- Features that could be added in the future
- Possible improvements (hardware/software)
- Scalability or real-world deployment ideas
- Integration with IoT, AI, or advanced sensors (if relevant)

Ideas should be **realistic and technically sound**.

7. Conclusion

[Slide-based presentation recommended]

Students should summarize:

- What was achieved
- Key learning outcomes
- Final system capabilities

End with a **clear closing statement**.

Submission Guideline for the Recommended Teams for Project Show: Upload the video to the **Google Drive link** that you provided while filling out the **Project Show registration form** by the **deadline announced by the Project Show Committee**.

Failure to submit within this deadline will result in your project not being accepted. Check whether **the visibility of the Google Drive folder is Public** (so that anyone with the link can be Viewer).